Howard Wang

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EDUCATION

Swarthmore College — Swarthmore, PA 2022 - 2026

- **B.A.** in Mathematics; **B.S.** in Engineering Swarthmore, PA
- **GPA:** 3.80/4.00
- **Relevant Courses:** Multivariable Calculus, Linear Algebra, PDEs, Probability, Stochastic Processes, Biomedical Data Analysis, Engineering Mechanics, Thermofluids, Physical Systems & Optics, Electrical Circuits.

PUBLICATION & CONFERENCES

- **Wang, H.**, Piovoso, M., Zurakowski, R.. A Spatiotemporal Model of T-Cell Population Dynamics within a Lymph Node [Poster] Biomedical Engineering Society Conference, San Diego, CA, 2025.
- **Wang, H.**, Moser, A. Frequency Domain Analysis of Infant and Adult EEG During Object Recognition. [Poster] Society for Neuroscience Conference, Washington, DC, 2024.
- **Wang, H.**, Miller, J., Towles, J. System to Measure Thumb-Tip Movement Produced by Muscles. [Poster] NCUR, Long Beach, CA, 2024.

AWARDS & HONORS

- $6,000 Allen and Naomi Schneider Summer Research Fund, Swarthmore College. Jun 2025
- $6,000 Tarble Summer Research Fellowship, Swarthmore College. Jun 2024
- $5,500 Surdna Summer Research Fellowship, Swarthmore College. Jun 2023
- $1,500 Society for Neuroscience Conference Travel Award, Swarthmore College & Sigma Xi. Oct 2024
- $1,000 NCUR Conference Travel Award, Sigma Xi & Provost's Office. Apr 2024
- $2,500 BMES Conference Travel Award, Sigma Xi & Provost's Office. Sep 2025
- 4x Centennial Academic Honor Roll 2023-2025

RESEARCH EXPERIENCES

Spatial Modeling of T Cell Dynamics in HIV Reservoir Formation Newark, DE

Dr. Ryan Zurakowski, Department of Biomedical Engineering, University of Delaware Sep 2024 - Present

- Constructed a PDE-based spatial framework to simulate CD4+/CD8+ T-cell and HIV reservoir dynamics under antiretroviral therapy, incorporating drug diffusion and inter-compartmental flux across blood, lymph, and follicular tissue.
- Captured spatial heterogeneity in infected and uninfected CD4+/CD8+ T-cell populations across 3D lymphoid architecture.
- Developed visualize and compare treatment outcomes across varying immune and drug parameters.
- Validated robustness by showing 95% MCMC credible intervals consistently encompassed observed T cell dynamics.

Data Analysis and System Design for Brain Computer Interface Research Remote

Ruten Inc. Aug 2025 – Present

- Analyzed macaque feeding behavior to reconstruct 3D gape cycles from high-speed video data.
- Developed a validation framework to assess computer-vision outputs, achieving spatial accuracy within ±5 mm per frame.

Frequency Domain Analysis of EEG in Infant and Adult Object Recognition Swarthmore, PA

Dr. Allan Moser & Dr. Benjamin Zinszer, Department of Engineering, Swarthmore College Jan 2023 – Jan 2025

- Applied Fast Fourier Transform to EEG data from infants and adults during object recognition tasks to extract 18 spectral features.
- Performed classification using SVM, achieving 96.60% accuracy based on theta/(alpha+beta) ratio features.
- Mapped coherence across parietal and occipital electrodes, identifying key developmental differences in visual processing.
- Shown results at Society for Neuroscience 2024; presented thesis on frequency-domain biomarkers of neural development.

Designing a Motion Capture System for Thumb-Tip Kinematics Swarthmore, PA
Dr. Joseph Towles, Department of Engineering, Swarthmore College Sep 2023 - May 2024

- Developed an opto-mechanical system to measure 3D thumb-tip trajectories in cadaveric specimens during muscle activation.
- Calibrated and validated motion capture system using digitized anatomical reference frames and mechanical rig testing.
- Analyzed variation in tip displacement across intrinsic thumb muscles, contributing to models of motor control and rehabilitation.
- Co-authored abstract and presented findings at NCUR 2024.

Molecular Dynamics Simulation of Tyrosine Phosphatase Binding Salt Lake City, Utah
Dr. Thomas Cheatham, College of Pharmacy May 2022 – Aug 2022

- Constructed all-atom protein models using Gaussian to evaluate ligand binding to tyrosine phosphatase.
- Conducted energy minimization and molecular dynamics to explore conformational shifts under thermal fluctuation.
- Interpreted hydrogen bonding patterns and residue-level interactions to assess active site accessibility for inhibitor design.

TEACHING EXPERIENCES

Grader Mechanics & Electrical Circuits & Computer Engineering & Stochastic Jan 2024 – May 2024

- Evaluated homework and problem sets for three core engineering courses (E6: Mechanics, E11: Electrical Circuits, E21: Computer Engineering, Math66: Numerical Methods & Stochastics, Physics3: Mechanics) with 120+ enrolled students.
- Provided accurate and timely feedback on conceptual and numerical problem-solving assignments.
- Collaborated with course instructors to clarify grading rubrics and adjust problem difficulty based on class performance.

EXTRACURRICULARS AND LEADERSHIP EXPERIENCES

Senior Admissions Fellow Swarthmore, PA
Swarthmore College Admissions Office May 2025 - Present

- Selected through a competitive application process to represent the college in interviewing prospective applicants.
- Conducted evaluative interviews and provided written reports used in the admissions decision process.
- Collaborated with admissions officers on outreach and programming for prospective students.

Team Vice Captain – Men's Cross Country Swarthmore, PA
Swarthmore College Varsity Athletics Mar 2022 - Present

- Led team warm-ups, organized travel logistics, and mentored first-year athletes on training and recovery practices.
- Helped foster a cohesive and motivated team environment through peer support and communication with coaches.

President – Swarthmore Calligraphy Club Swarthmore, PA
Swarthmore College Student Clubs Sep 2022 - Present

- Revived and led weekly workshops promoting East Asian calligraphy and ink-based visual art.
- Designed events to attract new members and collaborated with the Intercultural Center on programming.

Photo Editor – The Phoenix (Swarthmore Student Newspaper) & Swarthmore College Media Swarthmore, PA
Campus Publication & Social Media Sep 2022 – Present

- Managed a team of photographers and curated weekly image content for front-page and feature articles.
- Covered major campus events and theatrical productions with high-impact visual storytelling.

SKILLS & INTERESTS

- **Programming:** Python, Java, MATLAB, R, C/C++, LabView, Git
- **Technical Skills:** Machine learning, Signal processing, EEG analysis, ODE/PDE modeling
- **Art & Media:** Chinese calligraphy, oil color/watercolor painting, Documentary filmmaking (ultrarunning film with 20K+ views)
- **Languages:** English (fluent), Mandarin Chinese (native)
- **Interests:** Ultrarunning (6× 100-milers, 2:44 Marathoner), Photography